

Program – 10: Kubernetes Deployment and Service using Minikube

Aim

To deploy an Nginx application using a Kubernetes Deployment and expose it using a NodePort Service.

Requirements

- Ubuntu system
- Minikube
- kubectl
- deployment.yaml, service.yaml

Step 1: Start Minikube

minikube start

Step 2: Create Deployment

kubectl apply -f deployment.yaml

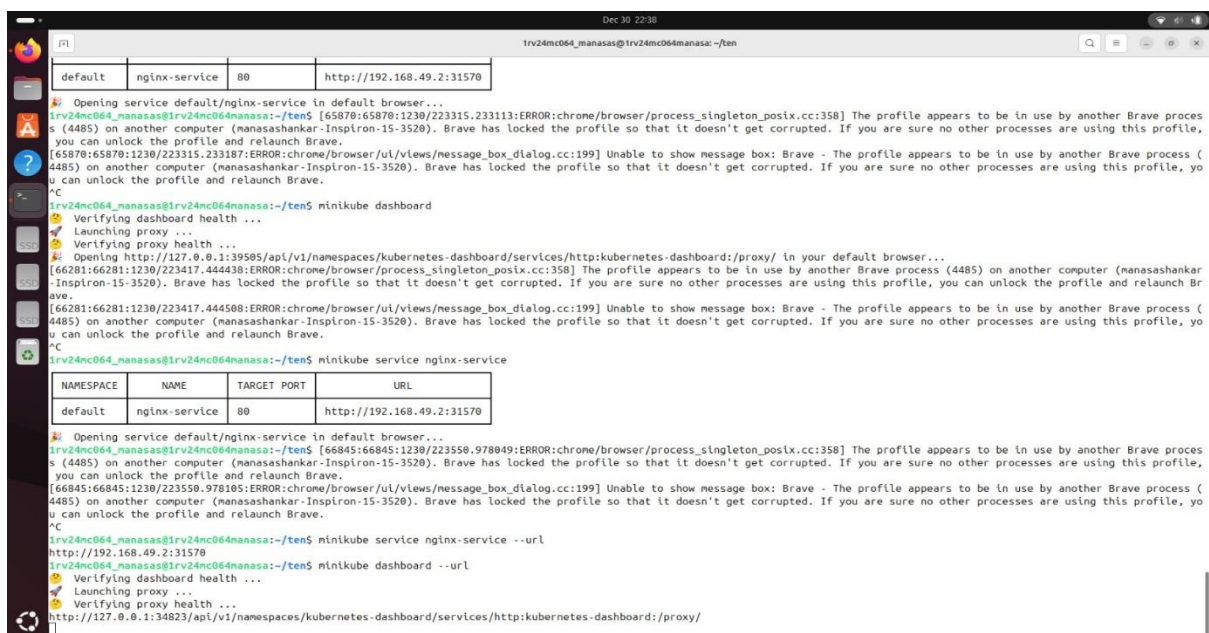
kubectl get deployments

kubectl get pods

Step 3: Create Service

kubectl apply -f service.yaml

kubectl get svc



```
1rv24mc064_manasas@1rv24mc064manasa: ~/ten
default      nginx-service  80      http://192.168.49.2:31570

$ kubectl apply -f deployment.yaml
deployment.apps/nginx-service created

$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-service  1/1     1            1           1m

$ kubectl get pods
NAME                               READY   STATUS    RESTARTS   AGE
nginx-service-5789f8b8-4444-4444-4444-4444  1/1     Running    0           1m

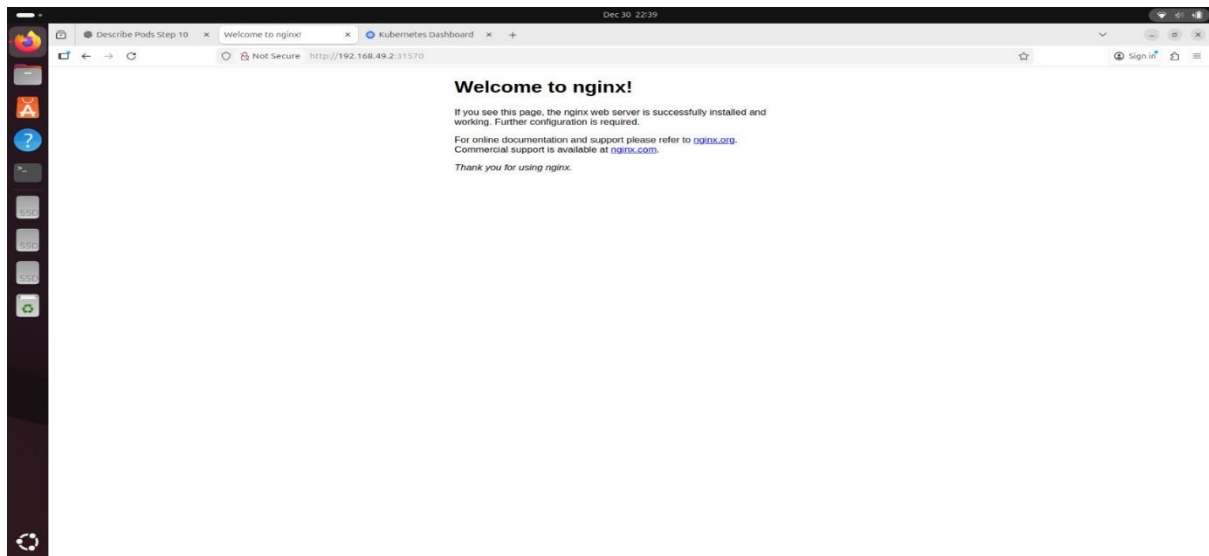
$ kubectl apply -f service.yaml
service/nginx-service created

$ kubectl get svc
NAME          TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          URL
nginx-service  NodePort    10.10.10.10      <none>         80 (TCP)         http://192.168.49.2:31570
```

Step 4: Access Application

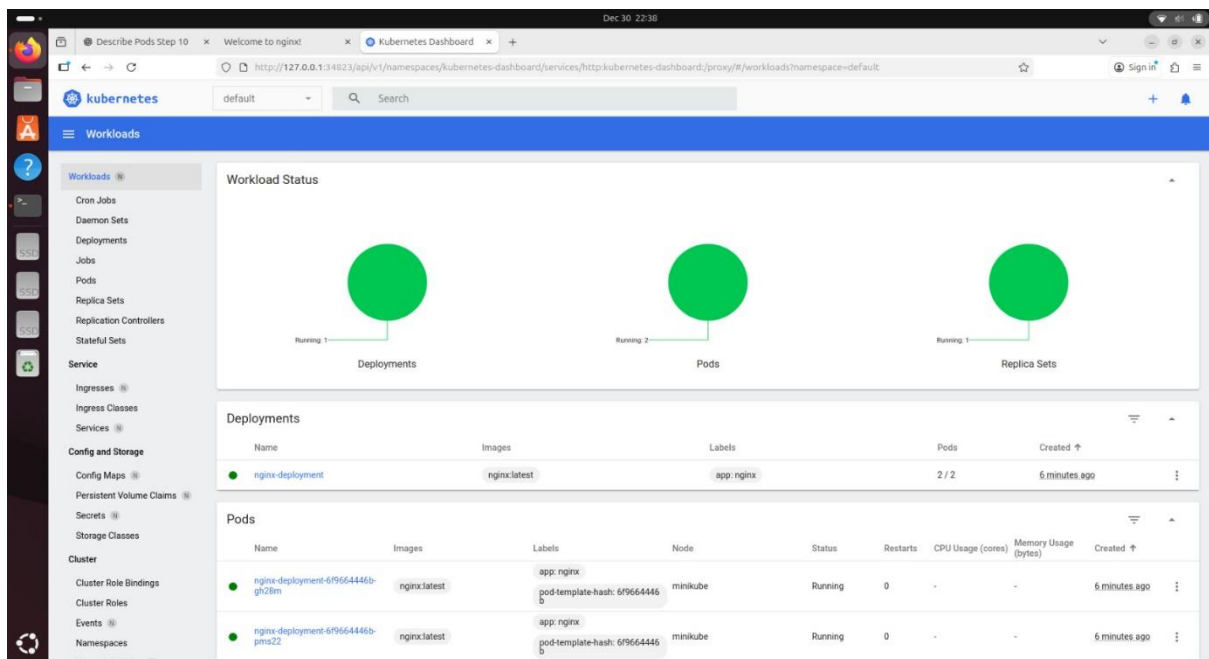
minikube service nginx-service --url

Open the URL in a browser to view the Nginx page.



Step 5: (Optional) Dashboard

minikube dashboard



Result

Nginx was deployed with two replicas and exposed using a NodePort service.